

EMMA KUUTTI

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SUMMARY

Master's student in Biomedical Engineering at Aalto University with a background in applied physics. Experienced in laboratory work, teaching assistance, and computational tools such as MATLAB and Python. Interested in AI, and data-driven approaches in healthcare.

EDUCATION

Aalto University

MSc (Tech), Biomedical Engineering | Minor: Engineering Physics | GPA 5/5

2026 – Present

BSc, Engineering Physics | Minor: Bio-IT | GPA 4.0/5

2023 – 2026

EXPERIENCE

Research Assistant | Helsingin yliopistollinen sairaala (HUS)

May 2026 – present

- Investigating the effect of sparse annotation strategies on segmentation accuracy using nnU-Net, one of the leading frameworks for medical image segmentation, in a clinically motivated research context.
- Training and evaluating deep learning models on Aalto University's Triton HPC cluster, gaining hands-on experience with large-scale scientific computing workflows.

Research Assistant | Department of Neuroscience and Biomedical Engineering, Aalto University

Jun 2025 – Oct 2025

- Conducted an extensive literature review within the Cellular Biophysics research group in Aalto University on electrophysiological measurement methods for retinal cells as part of Bachelor's thesis work
- Synthesized scientific findings into a structured thesis, developing skills in critical analysis, scientific writing, and data interpretation

Teaching Assistant | Department of Applied Physics, Aalto University

Sept 2024 – Feb 2026

- Guided students in Mechanics and Structure of Matter laboratories and led problem-solving sessions in Electromagnetism, supporting experimental work, data analysis, and understanding of complex physical concepts
- Instructed and mentored groups of 20+ students, strengthening communication, teaching, and teamwork skills in academic settings

SKILLS

Programming :

- Python** (numpy, scipy, pandas), **Machine learning** (sci-kit learn), **MATLAB** (data-analysis, simulations), **Scala**, **Jupyter Notebooks**, **HPC computing** (nnU_Net)
- Projects:** Please, have a look <https://emmakuutti.github.io/my-portfolio/>
- Tools:** Office Suite (Excel, Word, PowerPoint), LaTeX, Canva

LANGUAGES

Finnish: Native

English: Proficient

Swedish: Intermediate

LEADERSHIP & VOLUNTEERING

Scout Leader | Kaukupartio ry

Led and supervised groups of young scouts in weekly activities and camps

2019 – Present

Student representative | Life Sciences Program

Represented student perspectives in program-level discussions and decision-making

Jan 2026 – Present

Study Tutor | Guild of Physics

Mentored new students, supporting their transition into university studies

Sep 2025 – May 2026